

$\forall x, y \in \mathbb{R}, x < y$, and $\exists K: K < x < K + \eta$,
and $K' < y < K' + \eta$ ($K < K'$)
 $[x] = K < K' = [y]$

$$x < y \Rightarrow x + K < y + K' \dots \textcircled{II}$$

$$\Rightarrow K(x + K) < K'(y + K') \Rightarrow K(x + K)^K < K'(x + K)^{K'}$$

From (I) and (II) $\Rightarrow f$ is increasing.

Exercise 04:

① $f(x) = 3x - x^3 + 3x^7$

$D_f = \mathbb{R}$, $f(-x) = -f(x) \rightarrow$ odd

② $f(x) = \sqrt[8]{(1-x)^2} + \sqrt[8]{(1+x)^2}$

$D_f = \mathbb{R}$

$f_2(-x) = f(x) \rightarrow$ even.

$i f: f_2(x) = \sqrt[8]{(1-x)^2} - \sqrt[8]{(1+x)^2} \rightarrow$ odd

③ $f(x) = a^{\sin(x)} + a^{-\sin(x)}$ ($a > 0$)

$D_f = \mathbb{R}$

$f_3(-x) = f(x) \rightarrow$ even.

$f_3(x) = a^{\sin x} - a^{-\sin x} \rightarrow$ odd.

④ $f(x) = \ln\left(\frac{4-x}{4+x}\right)$ $\ln\left(\frac{4+x}{4-x}\right)$

$D_f =]-4, 4[$

$\rightarrow$ odd.

⑤ $f(x) = \ln(x + \sqrt{1+x^2})$

$D_f = \mathbb{R}$

$f(-x) = -f(x) \rightarrow$ odd.

Exercise 05:

① $\lim_{x \rightarrow -2} (x^2 - 1) = 3$

$\Leftrightarrow \forall \varepsilon > 0, \exists \delta(\varepsilon, 2) > 0, \forall x: |x+2| < \delta \Rightarrow$

$|f(x) - 3| < \varepsilon$

one has: $|f(x) - 3| = |x^2 - 4| = |x-2| \underbrace{|x+2|}$

let $x: |x - (-2)| < 1 \Leftrightarrow$

$-3 < x < -1$

$\Leftrightarrow -5 < x - 2 < -3$

$\Leftrightarrow |x - 2| < 5$

so: $|x+2| \leq \frac{\varepsilon}{5}$

It is sufficient to take:

$\delta(\varepsilon, -2) = \min\left\{1, \frac{\varepsilon}{5}\right\}$

② $\lim_{x \rightarrow -2} \frac{x^2 + x + 1}{1 - x} = -7 \Leftrightarrow$

$\forall \varepsilon > 0, \exists \delta(\varepsilon, 2) > 0, \forall x: |x - 2| < \delta \Rightarrow$

$|f(x) - (-7)| < \varepsilon$

one has:

$\left| \frac{x^2 + x + 1}{1 - x} + 7 \right| = \left| \frac{x^2 + x + 1 + 7 - 7x}{1 - x} \right|$

$= \left| \frac{x^2 - 6x + 8}{1 - x} \right| = \left| \frac{(x-2)(x-4)}{1-x} \right|$

$= \frac{|x-4|}{|1-x|} |x-2|$

Let $|x-2| < \frac{\eta}{2}$

$\Leftrightarrow \frac{3}{2} < x < \frac{5}{2} \Leftrightarrow \begin{cases} -2,5 < x-4 < -1,5 \\ -1,5 < 1-x < -0,5 \end{cases}$

$\Rightarrow \begin{cases} \Rightarrow |x-4| < 2,5 \\ \Rightarrow |1-x| > 0,5 \end{cases}$

$\downarrow$
 $\frac{1}{1,5} < \frac{1}{|1-x|} < \frac{1}{0,5} = 2$

So: $\frac{|x-4|}{|1-x|} \cdot |x-2| < \frac{2,5}{2} |x-2| \leq \varepsilon$

$\Rightarrow |x-2| < \frac{2\varepsilon}{2,5}$

$\delta = \min\left\{\frac{1}{2}, \frac{2\varepsilon}{2,5}\right\}$

ex ob:

$$a) \lim_{x \rightarrow 0} x \cos \frac{1}{x} = 0.$$

$$-1 < \cos \left(\frac{1}{x} \right) < 1$$

$$0 \swarrow \leftarrow -x < x \cos \left(\frac{1}{x} \right) < x \searrow \rightarrow 0$$

ex 068

a) $\lim_{x \rightarrow 0} \frac{L(x)}{x}$ doesn't exist.

For $x > 0$, then $L(x) = 0$ no.

$$\lim_{x \rightarrow 0^+} \frac{L(x)}{x} = 0$$

if $x < 0$, then $L(x) = -1$ so.

$$\lim_{x \rightarrow 0^-} \frac{L(x)}{x} = +\infty$$

c) $\lim_{x \rightarrow +\infty} (x(\sqrt{x^2+1} - \sqrt[3]{x^3+1})) = \frac{1}{2}$

$$= \lim_{x \rightarrow +\infty} \left[x \frac{(\sqrt{x^2+1}) - \sqrt[3]{x^3+1}}{\sqrt{x^2+1} + \sqrt[3]{x^3+1}} \right]$$

$$= \lim_{x \rightarrow +\infty} \left[x \frac{(x^2+1) - (x^3+1)^{2/3}}{\sqrt{x^2+1} + \sqrt[3]{x^3+1}} \right]$$

using the formula $A^3 - B^3 = (A-B)(A^2 + AB + B^2)$

where $A = \sqrt{x^2+1}$ and $B = \sqrt[3]{x^3+1}$

$$= \lim_{x \rightarrow +\infty} \left[x \frac{((x^2+1) - (x^3+1)^{2/3})((x^2+1)^2 + (x^2+1)\sqrt[3]{x^3+1} + (\sqrt[3]{x^3+1})^2)}{(\sqrt{x^2+1} + \sqrt[3]{x^3+1})(\quad)} \right]$$

$$= \lim_{x \rightarrow +\infty} \left[x \frac{(x^2+1)^3 - (x^3+1)^2}{(\quad)(\quad)} \right]$$

$$= \frac{1}{2}$$

2. mini

$$f) \lim_{x \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} \cos x\right)}{\sin(\sin x)} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \left[\frac{\cos\left(\frac{\pi}{2} \cos(x)\right)}{\frac{\pi}{2}(1 - \cos(x))} \cdot \frac{\sin(x)}{\sin(\sin(x))} \cdot \frac{\frac{\pi}{2}(1 - \cos(x))}{\sin(x)} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{\cos\left(\frac{\pi}{2} \cos(x)\right)}{\frac{\pi}{2} - \frac{\pi}{2} \cos(x)} \cdot \frac{1}{\frac{\sin(x)}{\sin(\sin(x))}} \cdot \frac{\frac{\pi}{2}(1 - \cos(x))(1 + \cos(x))}{\sin(x)(1 + \cos(x))} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{-\cos\left(\frac{\pi}{2} \cos(x)\right)}{\frac{\pi}{2} \cos(x) - \frac{\pi}{2}} \cdot \frac{1}{\frac{\sin(x)}{\sin(\sin(x))}} \cdot \frac{\frac{\pi}{2} \sin(x)}{1 + \cos(x)} \right]$$

$\downarrow$ $\downarrow$ $\downarrow$
 1 1 0

$$= 0$$

ex. of: $f(x) = \cos(\ln(x))$

f doesn't have limit at $+\infty$. Indeed, let

take $y_n = e^{n\pi}$

then: $f(y_n) = \cos(\ln(e^{n\pi})) = \cos(n\pi) = (-1)^n$

$$= \frac{1}{2}$$

sin

$$\begin{aligned} f) \lim_{k \rightarrow \infty} \frac{\cos\left(\frac{\pi}{2} \cos k\right)}{\sin(\sin k)} &= \frac{0}{0} \\ &= \lim_{k \rightarrow \infty} \left[\frac{\cos\left(\frac{\pi}{2} \cos(k)\right)}{\frac{\pi}{2}(1 - \cos(k))} \cdot \frac{\sin(k)}{\sin(\sin k)} \cdot \frac{\frac{\pi}{2}(1 - \cos(k))}{\sin(k)} \right] \\ &= \lim_{k \rightarrow \infty} \left[\frac{\cos\left(\frac{\pi}{2} \cos k\right)}{\frac{\pi}{2} - \frac{\pi}{2} \cos(k)} \cdot \frac{1}{\frac{\sin(\sin k)}{\sin k}} \cdot \frac{\frac{\pi}{2}(1 - \cos k)(1 + \cos k)}{\sin(k)(1 + \cos k)} \right] \\ &= \lim_{k \rightarrow \infty} \left[\frac{-\cos\left(\frac{\pi}{2} \cos(k)\right)}{\frac{\pi}{2} \cos k - \frac{\pi}{2}} \cdot \frac{1}{\frac{\sin(\sin k)}{\sin k}} \cdot \frac{\frac{\pi}{2} \sin(k)}{1 + \cos(k)} \right] \\ &= 0 \end{aligned}$$

ex 07: $f(x) = \cos(\ln(x))$

f doesn't have limit at $+\infty$. Indeed, we

take $y_n = e^{n\pi}$

then: $f(y_n) = \cos(\ln(e^{n\pi})) = \cos(n\pi) = (-1)^n$

ex 08:

a): $\lim_{k \rightarrow \infty} (x - \sqrt{x^2 - 2x}) = 1$

$$= \lim_{k \rightarrow \infty} \left(\frac{(x - \sqrt{x^2 - 2x})(x + \sqrt{x^2 - 2x})}{x + \sqrt{x^2 - 2x}} \right)$$

$$= \lim_{\kappa \rightarrow \infty} \frac{\kappa^2 - \kappa^2 + 2\kappa}{\kappa + \sqrt{\kappa^2 - 2\kappa}} = \lim_{\kappa \rightarrow \infty} \frac{2\kappa}{\kappa + \sqrt{\kappa^2 - 2\kappa}}$$

$$= \lim_{\kappa \rightarrow \infty} \frac{\kappa}{\kappa} \left(\frac{2}{1 + \sqrt{1 - \frac{2}{\kappa}}} \right) = 1.$$

b) $\lim_{\kappa \rightarrow \infty} \left(\frac{\sqrt{\kappa}}{\sqrt{\kappa + \sqrt{\kappa + \sqrt{\kappa}}}} \right) = 1$ because

$$= \lim_{\kappa \rightarrow \infty} \left(\frac{\sqrt{\kappa}}{\sqrt{\kappa + \sqrt{\kappa^2 \frac{\kappa}{\kappa^2} + \kappa^2 \sqrt{\frac{\kappa}{\kappa^4}}}}} \right)$$

$$= \lim_{\kappa \rightarrow \infty} \frac{\sqrt{\kappa}}{\sqrt{\kappa + \sqrt{\kappa^2 \left(\frac{1}{\kappa} + \sqrt{\frac{1}{\kappa^3}} \right)}}}$$

$$= \lim_{\kappa \rightarrow \infty} \left[\frac{\sqrt{\kappa}}{\sqrt{\kappa} \left(\sqrt{1 + \sqrt{\frac{1}{\kappa} + \sqrt{\frac{1}{\kappa^3}}}} \right)} \right] = 1.$$

$$\lim_{x \rightarrow 0} \frac{x - L(x)}{\sqrt{|x|}}$$

$$\lim_{x \rightarrow 0} \frac{x - L(x)}{\sqrt{x}}$$

$x \rightarrow 0^+$, then $x \in]0, 1[$ then $L(x) = 0$

$$\lim_{x \rightarrow 0^+} \frac{x}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \sqrt{x} = 0$$

If $x \rightarrow 0^-$ then $x \in [-1, 0[$, then $L(x) = -1$

$$\lim_{x \rightarrow 0} \frac{x+1}{\sqrt{-x}} = +\infty$$

$$3) \lim_{x \rightarrow 0} \frac{L(\frac{1}{x}) + x}{L(\frac{1}{x}) - x} = \lim_{x \rightarrow 0} \frac{\frac{L(\frac{1}{x})}{\frac{1}{x}} + \frac{x}{\frac{1}{x}}}{\frac{L(\frac{1}{x})}{\frac{1}{x}} - \frac{x}{\frac{1}{x}}}$$

$$= \lim_{x \rightarrow 0} \frac{1+x^2}{1-x^2} = 1$$

$\exists \epsilon > 0, \delta > 0$

a) 1. $x > 0$

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{x^\alpha} = +\infty$$

$$\lim_{x \rightarrow +\infty} \frac{e^{\ln(x^\alpha)}}{e^{\ln x^\alpha}} = \lim_{x \rightarrow +\infty} \frac{e^{\alpha \ln(x)}}{e^{\ln x^\alpha}}$$

Since $\alpha > 1, \alpha > 0$

$$= \frac{x^{\alpha \ln(x)} - \ln(x)}{e^{\ln x^\alpha}} = \frac{x^{\alpha \ln(x)} - \ln(x)}{e^{\alpha \ln(x)}} = +\infty$$

$$\lim_{x \rightarrow +\infty} \frac{\ln(x)}{x^\alpha} = \lim_{x \rightarrow +\infty} \frac{\ln(\ln(x))}{e^{\ln x^\alpha}} = \lim_{x \rightarrow +\infty} \frac{\ln(\ln(x))}{x^\alpha \ln(x)}$$

$$\ln(\ln(x)) - \alpha \ln(x)$$

but, $\frac{1}{x} = t$

$$\lim_{t \rightarrow 0} \frac{L(t) + \frac{1}{t}}{L(t) - \frac{1}{t}} = \lim_{t \rightarrow 0} \frac{1 + \frac{1}{tL(t)}}{1 - \frac{1}{tL(t)}} = 1$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\sqrt{x + \sqrt{x + \sqrt{x}}}} &= \lim_{x \rightarrow \infty} \sqrt{\frac{x}{x + \sqrt{x + \sqrt{x}}}} = \sqrt{\frac{1}{1 + \frac{\sqrt{x + \sqrt{x}}}{x}}} \\ &= \lim_{x \rightarrow \infty} \sqrt{\frac{1}{1 + \frac{\sqrt{x + \sqrt{x}}}{x^2}}} = 1 \end{aligned}$$

Method $\lim_{x \rightarrow \infty} \frac{x^2}{(x + \sqrt{x + \sqrt{x}})^2} = \sqrt{\frac{x^2}{x^2 + x + \sqrt{x + \sqrt{x}}}} = \sqrt{1} = 1$

$$\lim_{x \rightarrow \infty} \left(1 + \sin^2 \frac{1}{x} \right)$$

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1 \Rightarrow 0 \leq \sin^2 \frac{1}{x} \leq 1.$$

$$\Rightarrow 1 \leq \sin^2 \frac{1}{x} + 1 \leq 2.$$

for $0 < x < 2$.

$$\lim_{x \rightarrow \infty} (x) < \infty.$$

$$\text{By } 1 \leq \left(1 + \sin^2 \frac{1}{x} \right) h(x) \leq h(x).$$

$$\lim_{x \rightarrow \infty} h(x) = -\infty.$$

$$\text{So } \lim_{x \rightarrow \infty} \left(1 + \sin^2 \frac{1}{x} \right) \cdot h(x) = -\infty.$$

e

(x)

0

$$f(x)$$

$x \rightarrow 0$

$$f(x) = \frac{1}{x} f\left(\frac{1}{x}\right)$$

$x \rightarrow 0$

$$f(x) = \frac{f(x)}{0}$$

$x \rightarrow 0$

exercise 10:

$$f \in C[a, b] \Rightarrow |f| \in C[a, b].$$

$$\forall \varepsilon > 0, \exists \delta(\varepsilon) > 0, \forall x \in U(x_0) \cap \{x_0\}.$$

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

using the converse triangle inequality we get:

$$||f(x)| - |f(x_0)|| \leq |f(x) - f(x_0)| < \varepsilon.$$

Hence $|f| \in C([a, b])$.

• the converse of implication isn't true,

Indeed, we define:

$$f(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0 \end{cases}$$

we have:

$$|f| \in C(\mathbb{R}) \text{ but:}$$

$$f \notin C(\mathbb{R}).$$

Ex 11:

1/

It's clear that f is continuous on $\mathbb{R}/\mathbb{Z}$
on $\mathbb{Z}$:

Let $k \in \mathbb{Z}$

$$\underline{n \leq k: [n] = k-1}$$

$$\underline{n \geq k: [n] = k}$$

$$\lim_{n \rightarrow k} f(n) = \lim_{n \rightarrow k} [k + \sqrt{n-k}] = k$$

$$\lim_{n \rightarrow k} f(n) = \lim_{n \rightarrow k} [k-1 + \sqrt{n-k+1}] = k$$

$$f(k) = k + \sqrt{k-k} = k$$

f is continuous on $\mathbb{Z}$
 $\Rightarrow f$ is " " on $\mathbb{R}$.

2/ g is continuous on $\mathbb{R}/\mathbb{Z}$
on $\mathbb{Z}$:

$$n \leq k: [n] = k-1$$

$$n \geq k: [n] = k$$

$$\lim_{n \rightarrow k} g(n) = \lim_{n \rightarrow k} \left(k-1 - (n-k+1)^2 \right) = k-1$$

$$\lim_{x \rightarrow k} g(x) = \lim_{x \rightarrow k} (k - (x - k)^2) = k$$

$$g(k) = k$$

g is not continuous on $\mathbb{Z}$
 $\Rightarrow g$ is continuous on $\mathbb{R} \setminus \mathbb{Z}$.

exercise 12:

$$f(x) = \begin{cases} x^2 - 1 & \text{if } x \text{ is irrational} \\ 0 & \text{if } x \text{ is rational} \end{cases}$$

$x_0 \in \mathbb{Q}$ $x \in \mathbb{Q}$.

$$\lim_{\substack{x \rightarrow x_0 \\ x_0 \in \mathbb{Q}^c}} f(x) = f(x_0) \quad A \subset \mathbb{R}$$

$$\lim_{\substack{x \rightarrow x_0 \\ x_0 \in \mathbb{Q}}} f(x) = f(x_0) \quad \overline{A} = \mathbb{R}$$

$\Updownarrow$
 $x_0 \in \mathbb{Q} \subset \mathbb{R}$

$$\forall x_0 \in \mathbb{R}; \exists (x_n) \subset A$$

$x \longrightarrow x_0$

$$\overline{\mathbb{Q}} = \mathbb{R}$$

$x_0 \in \mathbb{Q}^c \subset \mathbb{R}$

$$\exists (x_n) \subset \mathbb{Q}$$

$n \longrightarrow +\infty$
 $x_n \longrightarrow x_0$

Exercise 13:

1. $f(x) = \frac{x}{4-x^2}$ $I = [-1, 1]$

f is continuous on $[-1, 1]$, so f is uniformly continuous on $[-1, 1]$.

2. $f_3(x) = \frac{\sin x}{x}$, $I = [0, \pi]$

Since f_3 is continuously extend at 0, so f_3 is continuous on $[0, \pi]$.

consequently, $f_3 \in UC([0, \pi])$.

3. $\cos(x) = 2 \sin(x) - \cos(x)$.

f_n is uniformly continuous, because the sum of the uniformly continuous functions.

• $f_2(x) = \ln(x)$

using sequential characterization, take

$u_n = e^{-n}$, and $v_n = 2e^{-n}$.

So: $|u_n - v_n| \xrightarrow{n \rightarrow \infty} 0$ but $|f(u_n) - f(v_n)| \xrightarrow{n \rightarrow \infty} \ln(2)$.

Hence f_2 isn't uniformly continuous.

• $f_5(x) = \frac{1}{x}$; $I =]0, 1[, 1[$

f_5 is not uniformly continuous

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f is continuous, non constant
periodic
then it admits a primitive

By contradiction: suppose that
 f doesn't admit a last period
let (T_n) be a sequence of period

$$\lim_{n \rightarrow \infty} T_n = 0$$

let $x_0 \in \mathbb{R}$, f is continuous at $x_0 \Leftrightarrow$

$\forall \epsilon > 0, \exists \delta (E, x_0) > 0, \forall x:$

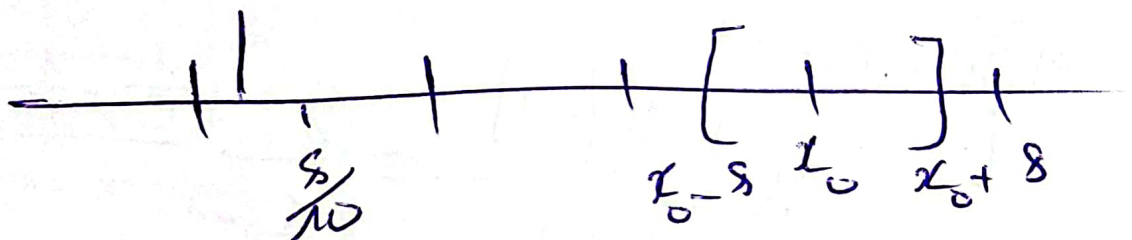
$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \epsilon$$

Since $T_n \rightarrow 0, \exists n_0 \in \mathbb{N}:$

$$0 < T_{n_0} < \frac{\delta}{10} \Rightarrow \exists k \in \mathbb{Z}$$

$$kT_{n_0} \in [x_0 - \delta, x_0 + \delta]$$

T_{n_0}



$$|f(0) - f(x_0)| = |f(kT_{n_0}) - f(x_0)| < \epsilon$$

and since ϵ is any given, so

$$|f(0) - f(x_0)| = 0 \text{ for } x_0 \in \mathbb{R}$$

f is constant. Contradiction.

$T_n \rightarrow 0$

x^2

if $|x| < 1$

6. 1. for all $x, y \in [0, a]$ we have

$$\begin{aligned} |\sin(x^2) - \sin(y^2)| &\leq |x^2 - y^2| = |x + y| |x - y| \\ &\leq (|x| + |y|) |x - y| \\ &\leq 2a |x - y|. \end{aligned} \quad \mathbf{1 \text{ point}}$$

This shows that f is lipshitzian on $[0, a]$. Therefore, f is uniformly continuous on $[0, a]$.

2. Let for $n \in \mathbb{N}$, $x_n = \sqrt{n\pi + \frac{\pi}{2}}$ and $y_n = \sqrt{n\pi}$. Then for all $n \in \mathbb{N}$ we have

$$|x_n - y_n| = \sqrt{n\pi + \frac{\pi}{2}} - \sqrt{n\pi} = \frac{\pi}{2(\sqrt{n\pi + \frac{\pi}{2}} + \sqrt{n\pi})} \leq \frac{4}{4\sqrt{n\pi}} = \frac{1}{\sqrt{n\pi}}$$

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and

$$\begin{aligned} |\sin(x_n^2) - \sin(y_n^2)| &= 2 \left| \sin\left(\frac{x_n^2 - y_n^2}{2}\right) \right| \left| \cos\left(\frac{x_n^2 + y_n^2}{2}\right) \right| \\ &= 2 \sin\left(\frac{\pi}{4}\right) \left| \cos\left(n\pi + \frac{\pi}{4}\right) \right| \\ &= 2 \sin\left(\frac{\pi}{4}\right) \left| \cos(n\pi) \cos\left(\frac{\pi}{4}\right) - \sin(n\pi) \sin\left(\frac{\pi}{4}\right) \right| \\ &= 2 \left(\frac{\sqrt{2}}{2}\right) \left| (-1)^n \left(\frac{\sqrt{2}}{2}\right) \right| \\ &= 1. \end{aligned} \quad \mathbf{2 \text{ points}}$$

Therefore, taking $\epsilon_0 = 1$, for any $\delta > 0$ we have $|x_{n_k} - y_{n_k}| \leq \delta$, where $n_k = E\left(\frac{1}{\epsilon_0 \delta}\right) + 1$, and $|f(x_{n_k}) - f(y_{n_k})| = 1$. This shows that f is not uniformly continuous on $[0, +\infty)$.

7. Let $\epsilon > 0$ be such that

$$(k - \epsilon) > 1 > (l + \epsilon).$$

One can choose $\epsilon = \min\left(\frac{k-1}{4}, \frac{1-l}{4}\right)$, in this case we have

$$\begin{aligned} 1 &= \frac{3+1}{4} > \frac{3l+1}{4} = l + \frac{1-l}{4} \geq l + \epsilon \text{ and} \\ 1 &= \frac{3+1}{4} < \frac{3k+1}{4} = k - \frac{k-1}{4} \leq k - \epsilon. \end{aligned} \quad \mathbf{1 \text{ point}}$$

From the definition of the limit we deduce that there is $A > 0, B > 0$ such that

$$\begin{aligned} F(x) &\geq (k - \epsilon)x \text{ for all } x \geq B, \text{ and} \\ F(x) &\leq (l + \epsilon)x \text{ for all } 0 < x \leq A. \end{aligned}$$

Consider the function $G : [A, B] \rightarrow \mathbb{R}$ with $G(x) = F(x) - x$ for all $x \in [A, B]$. Clearly G is continuous and we have

$$\begin{aligned} G(A) &= F(A) - A \leq (l + \epsilon)A - A = ((l + \epsilon) - 1)A < 0, \\ G(B) &= F(B) - B \geq (k - \epsilon)B - B = ((k - \epsilon) - 1)B > 0. \end{aligned} \quad \mathbf{1 \text{ point}}$$

By Intermediate Value Theorem, there exists $c \in (A, B)$ such that $G(c) = 0$, that is $F(c) = c$. This shows that the equation $F(x) = x$ admits a solution. $\mathbf{1 \text{ point}}$